Class 31

Incomplete information games (Harsanyi)

$$\langle N, (T_i)_{i \in N}, p, S, (s_t)_{t \in \times_{i \in N} T_i} \rangle$$

For a cards game example

- $N = \{1, 2, 3, 4\}$
- T_i = 13 cards that player i may get

$$|T_i| = \binom{52}{13}$$

- $p(t) = \begin{cases} \frac{1}{\binom{52}{13}\binom{39}{13}\binom{26}{13}} & \text{if } t \text{ is a case dissent} \\ 0 & \text{O.W.} \end{cases}$

$$T = T_1 \times T_2 \times T_3 \times T_4$$

$$|T| = \left[\binom{52}{13}\right]^4$$

- S = a collection of games for each type profile
- S_t: game corresponding to types profile t. where the action set of player i depends on t_i only.

How the game proceeds

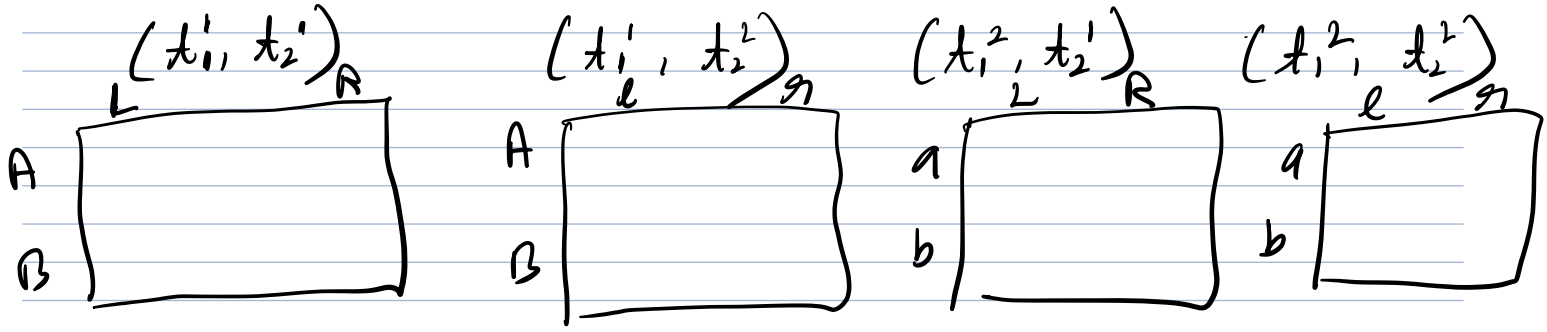
- Nature selects a type profile t according to p.
- All the players observe their own type only

- Players play an action from the set of actions available at his type. i.e, Player i plays an action from $a_i \in A_i(t_i)$

- At the end, the players get their utilities which depends on the action vectors $a = (a_1, \ldots, a_n)$ and the type profile $t = (t_1, \ldots, t_n)$

$N = \{1, 2\}$

$T_1 = \{t_1^1, t_1^2\} \quad T_2 = \{t_2^1, t_2^2\}$



p owns these four type profiles.

(t_1^1, t_2^1)

$s_1 : \{t_1^1, t_1^2\} \to A_1(t_1^1) \cup A_1(t_1^2)$

where $s_1(t_1^1) \in A_1(t_1^1)$

and

$s_1(t_1^2) \in A_1(t_1^2)$

The man has one type t_1 and the woman has two types (y, n)

y:

	F	B
F	2,1	0,0
B	0,0	1,2

n:

	F	B
F	2,0	0,2
B	0,1	1,0

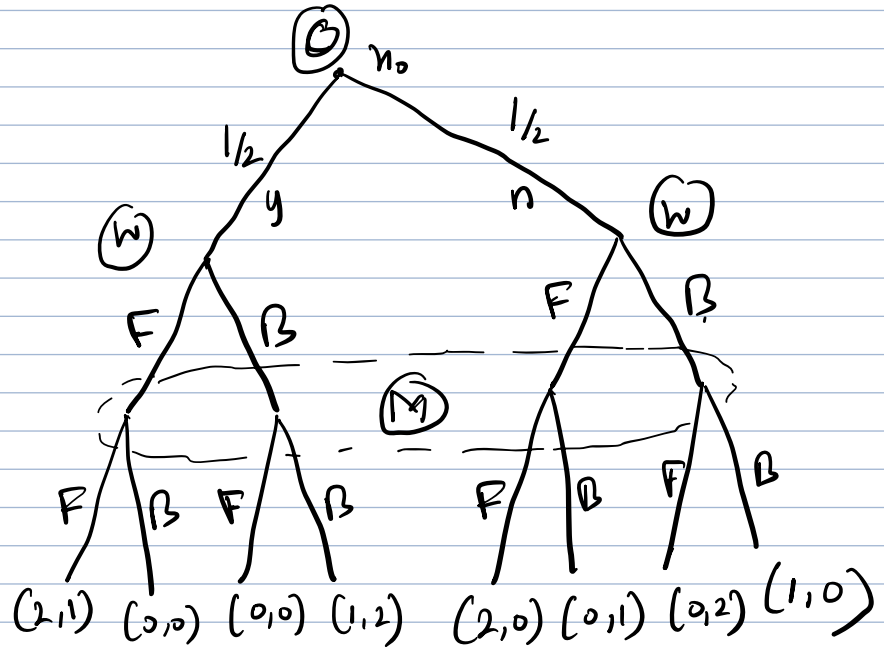
The man attaches $1/2$ probability to both games and the woman knows this.

$$p(1,y) = p(1,n) = 1/2$$

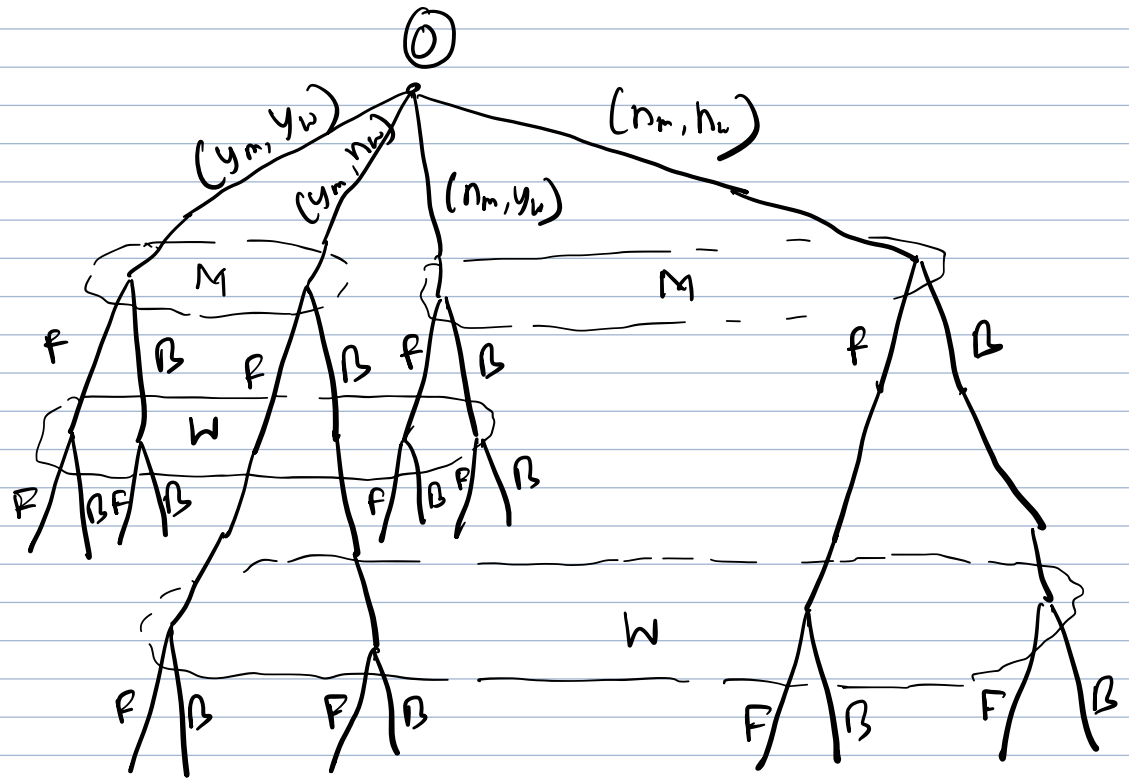
Player i after realizing his type t_i can compute

$$p(t_{-i} / t_i) \quad \text{for every } t_{-i} \in T_{-i}$$

$p(t_{-i} / t_i)$ is Player i's belief about the types of the other players being t_{-i}.



(y_m, y_w) (y_m, n_w) (n_m, y_w) (n_m, n_w)



Theorem

Every (Harsanyi) game of incomplete information can be described as an extensive form game with chance moves and information sets.